Foundations of Machine Learning

Explainable Artificial Intelligence

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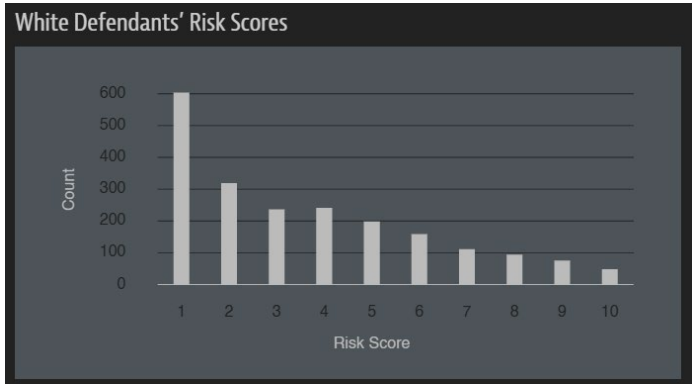
Why explainability? – Some examples

- Models can learn unfair and **discriminatory** patterns.
- Opaque models make it **hard to identify errors**.
- Models can make **mistakes**. Are they accountable?
- Models can perpetuate **historical biases**.
- Models can be **fooled**.
- …

COMPAS Score Case – Predicting recidivism

- Generates a risk score that predicts the **likelihood** of **future criminal activity**
- The algorithm is a "**black box**," making it difficult to identify the **factors considered** when classifying someone as high or low risk for reoffending.

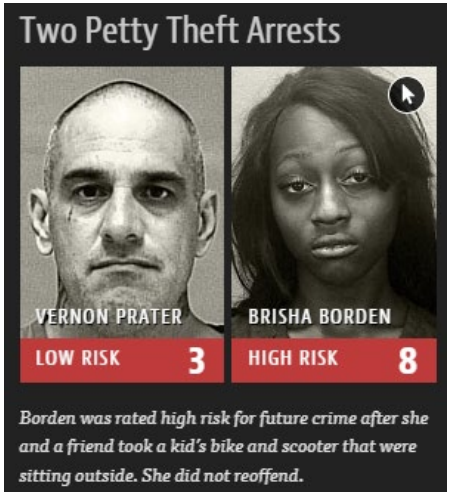


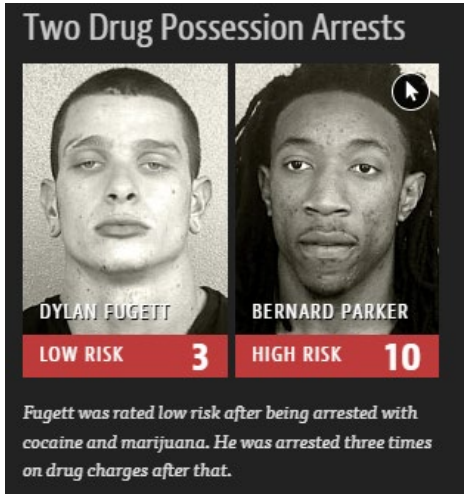
These charts show that scores for white defendants were skewed toward lower-risk categories. Scores for black defendants were not.

COMPAS Score Case – Predicting recidivism

- The algorithm wrongly assigned **African-American defendants with a high risk of recidivism**. The false positives are at almost twice the rate of white defendants

- **White defendants were mislabeled as low-risk** more often than black defendants





Source:

- https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing
- https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm

Air Canada – Fake refund policy

- A Canadian customer reportedly asked the **chatbot** for clarification on **refunding** and the **chatbot provided fake information** on the refund policy.
- The airline argued that "**the chatbot is a separate legal entity that is responsible for its own actions**"
- The court sided with the customer, forcing the company to issue the passenger a **partial refund** and pay for his court fees.

Air Canada Must Honor a Fake Refund Policy Created by Its Chatbot, Court Says

The airline argued that the chatbot should be responsible for its own actions.

By Emily Price Updated February 18, 2024

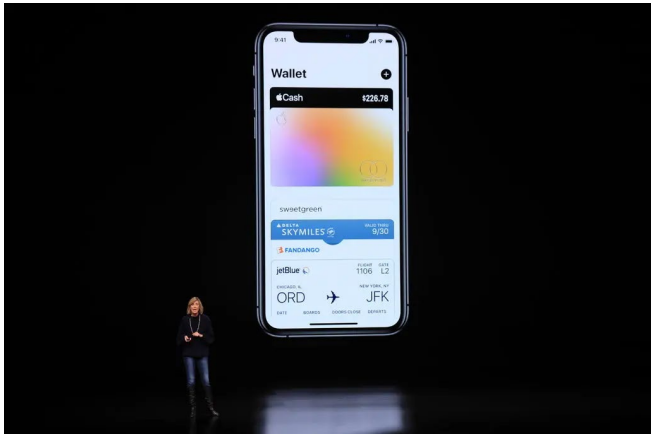
Source:

- https://www.pcmag.com/news/air-canada-must-honor-a-fake-refund-policy-created-by-its-chatbot-court

Models can be biased – Apple

- A husband and wife applied for an **Apple Card.** Despite the wife having a **better credit score** and other favorable factors, the husband **was granted a credit limit 20 times higher than hers.**

- The customer service employees were **unable to explain why the algorithm had designated her to be less creditworthy.**



Apple Card Investigated After Gender Discrimination Complaints

A prominent software developer said on Twitter that the credit card was "sexist" against women applying for credit.

Source:

- https://www.nytimes.com/2019/11/10/business/Apple-credit-card-investigation.html
- https://www.theverge.com/2021/3/23/22347127/goldman-sachs-apple-card-no-gender-discrimination

Models can be biased – Amazon

- Amazon decided to shut down its **experimental artificial intelligence** recruiting tool after discovering it **discriminated against women**.
- The algorithm learned to **systematically downgrade women's resumes for technical positions.**
- The algorithm was trained on all resumes submitted over a ten-year period to identify the best candidates. **The algorithm recognized male dominance and incorrectly associated it with success.**

Insight - Amazon scraps secret AI recruiting tool that showed bias against women

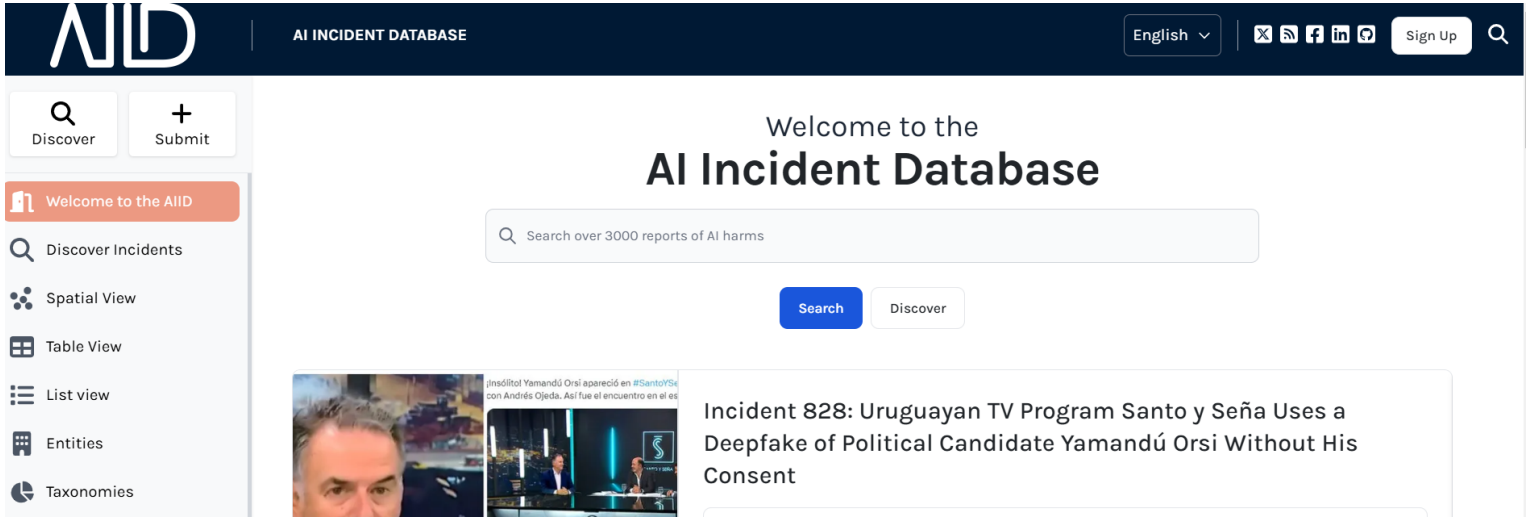
By Jeffrey Dastin

Source:

- https://www.reuters.com/article/us-amazon-com-jobs-automation-insight/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK08G/

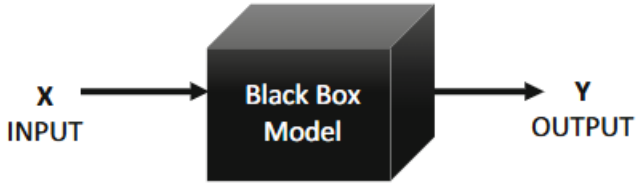
AI incidents database

- AI incidents database that registers AI failures and incidents.
- https://incidentdatabase.ai/



Black-Box problem

- We have a black box when we observe the set of outputs for a given set of inputs to a model, **without knowledge or understanding of its internal workings.**
- Unlike mathematical models that have inherent structure, **machine learning models can learn the mapping of inputs to outputs directly from the data.**
- For some models like decision trees, this **mapping is easily discernible.**
- For others like random forests or **deep learning** models, it becomes **next to impossible to understand how predictions are made.**



How do we know if and why the model predictions may be wrong?

The right to an explanation

"In any case, such processing should be subject to suitable safeguards, which should include specific information to the data subject and the right to obtain human intervention, to express his or her point of view, ***to obtain an explanation of the decision reached after such assessment and to challenge the decision."***

Recital 71 - GDPR

Transparency

*"The principle of **transparency** requires that any information addressed to the public or to the data subject be **concise**, **easily accessible and easy to understand**, and that clear and plain language and, additionally, where appropriate, **visualisation** be used."*

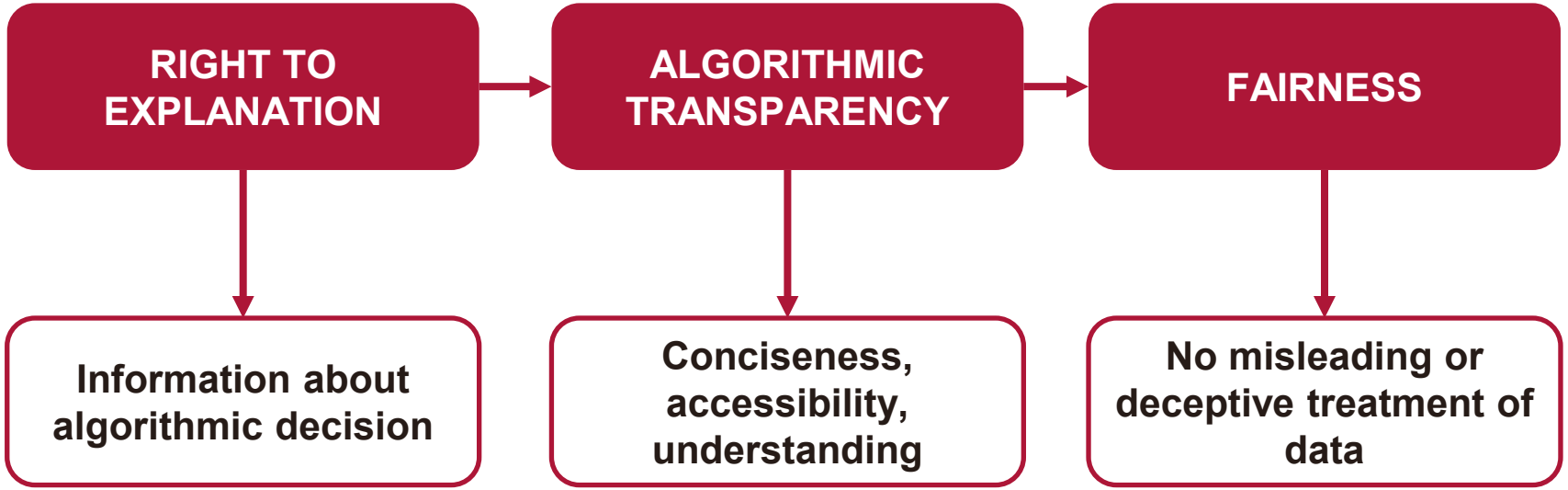
Recital 58 - GDPR

Fairness

*"The principles of **fair** and **transparent** processing require that the data subject be **informed of the existence of the processing operation and its purposes**. The controller should provide the data subject with any further information necessary to **ensure fair and transparent processing** taking into account the specific circumstances and context in which the personal data are processed."*

Recital 60 - GDPR

Fairness and Transparency

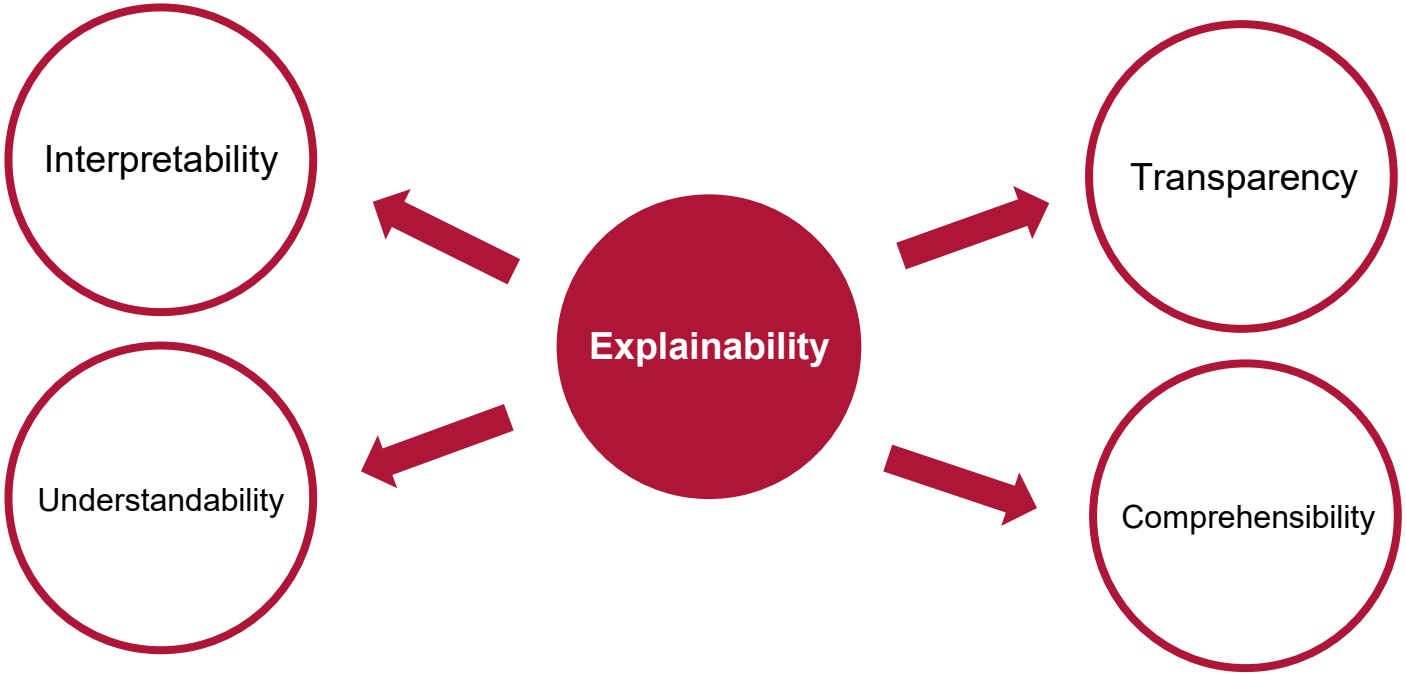


What does it mean to be fair?

- Employ appropriate **mathematical** or **statistical** procedures for the profiling.
- Implement technical and organizational measures appropriate to ensure that **factors that result in inaccuracies in personal data are corrected and the risk of errors is minimized.**
- **Secure personal data** in a manner that takes account of the potential risks involved for the interests and rights of the data subject
- **Prevent discriminatory effects on natural persons based on racial or ethnic origin, political opinion, religion, or beliefs**…

Recital 71 - GDPR

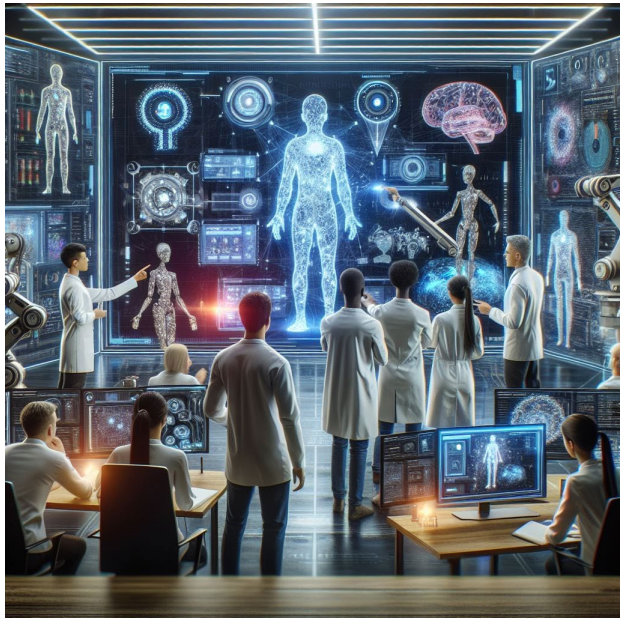
Explanation as a non-monolithic concept



Explainable AI (XAI)

- **Explainable AI (XAI)** seeks to provide us with insights into the decision-making ability of an AI system. It helps us to **understand** how, when, and **why** predictions are made [1].

- **XAI** is the ability of AI systems to provide **clear** and **understandable explanations** for their actions and decisions. Its central goal is to make the behavior of these systems **understandable** to **humans** by elucidating the **underlying mechanisms** of their decision-making processes [2].



Understandability

- **Understandability** (or equivalently, intelligibility) denotes the characteristic of a model to make a human **understand** its function – how the model works – without any need for explaining its internal structure or the algorithmic means by which the model processes data internally [3].

- **Understandability** is the notion that, to be useful, the underlying function of an AI model must be understandable to **humans**. The concept of understandability, also known as **intelligibility**, is the property of the overall model to be understandable **without any need for details and explanation** [2].

Interpretability

- **Interpretability**, often used interchangeably with explainability, is the **ability to explain or provide meaning to model predictions**. In particular, the goal of interpretability is **to describe the structure of a model in a fashion easily understandable by humans** [2].

- **Interpretability** refers to the degree of **human comprehensibility** of a given “black box” model or decision. Poorly interpretable models are **opaque** in the sense that when presented with the resulting decision, rarely does one have any concrete sense of how or **why a particular classification has been arrived at from inputs** [1].

What is a (good) explanation?

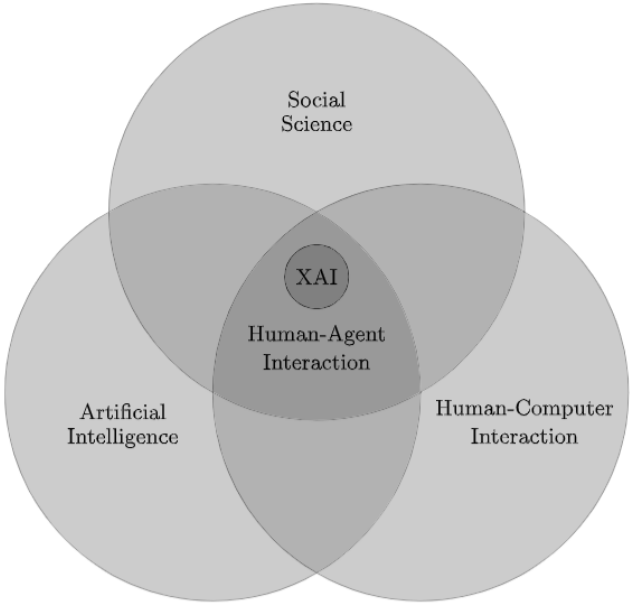
- Explanation answers to why questions:
  - Why did not the treatment work on the patient?
  - Why was my loan rejected?
  - Why my application did not pass the selection?

- There is a difference between explaining **good** and explaining **bad** and the act of explaining could be more or less satisfactory.

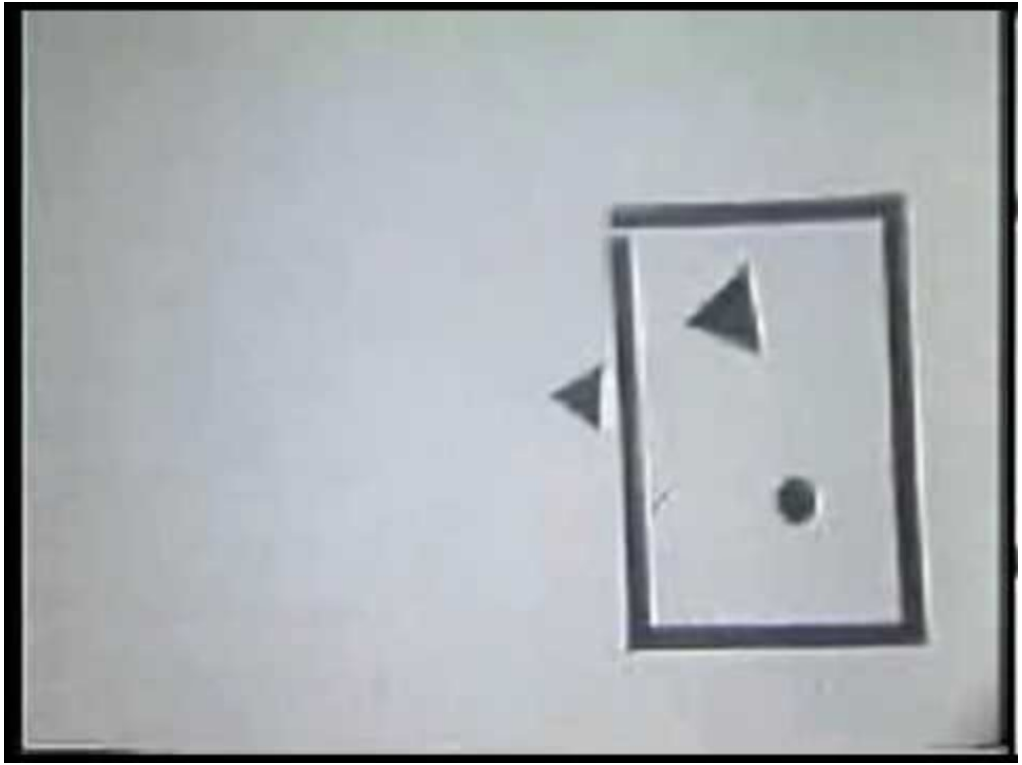
Good explanations – insights from social sciences

- What do **humans** consider a «good» explanation?

- Humanities can help us to find out: **explanations are a social interaction** between the **explainer** and the **explainee** and the social context has an influence.



How do people explain behavior?



Good explanations – insights from social sciences

- Explanations are **contrastive** [4].
- Explanatuons are **selected** (in a biased manner) [4].
- Probabilities don’t **matter** [4].
- Explanations are **social** [4].
- Explanations focus on the **abnormal** [5].

What does it mean for XAI?

CONTRASTIVE	Humans do not want a complete explanation for a prediction but want to compare what were the differences to another instance's prediction.
SELECTED	The explanation should be informative, but as short as the previous requisite allows.
PROBABILITIES DON'T MATTER	Seek for the causal nexus, rather than simple correlation.
SOCIAL	It should be paid attention to the target audience and the social environment of the machine learning application.
FOCUS ON ABNORMAL	If one of the input features for a prediction was abnormal and the feature influenced the prediction, it should be included in the explanation.

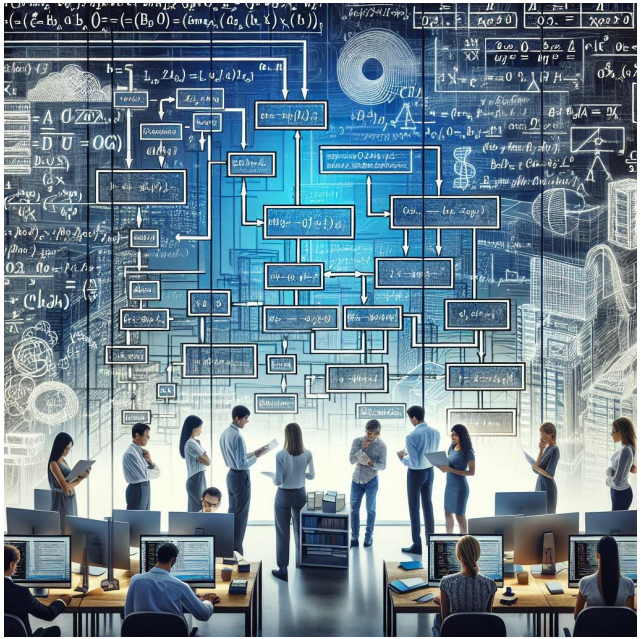
Desiderata from philosophy

- Philosophy of science has studied **scientific explanation** for centuries, from Aristotle to modern philosophy.

- Different principles exist of **«bona fide» explanations that can be useful also to XAI** [6]**.**
  - Factuality;
  - Similarity;
  - Exactness;
  - Fruitfulness;
  - Simplicity.

Properties of explanation methods

- Expressive Power;
- Translucency;
- Portability;
- Algorithmic Complexity.



Properties of individual explanations

- Accuracy;
- Fidelty;
- Consistency;
- Stability;
- Comprehensibility;
- Certainity;
- Degree of importance;
- Novelty;
- Representativeness.

Trade-offs

COMPLETENESS → INTERPRETABILITY

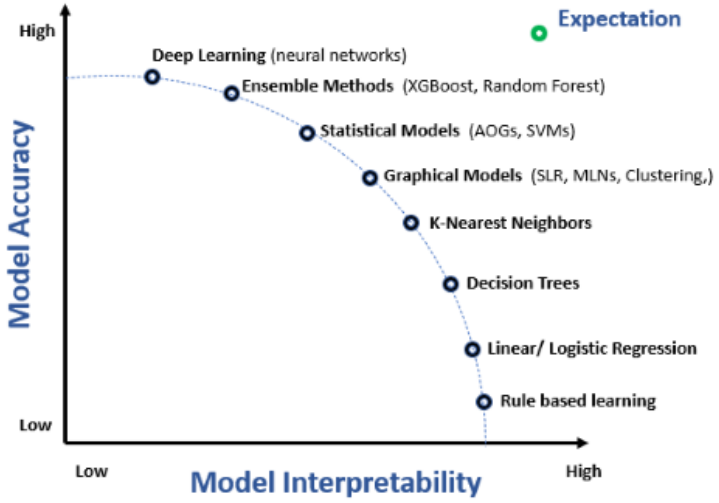
EFFICACY → PRIVACY

HUMAN EXPLANATIONS → ACCURACY

Completeness vs Interpretability

- Simple methods are **inherently interpretable** in that they are **self-explanatory by construction** and can provide useful explanations directly by inspection. However, these methods apply well to a **very limited set of problems** in the real world.

- **Deep neural networks** can approximate any continuous nonlinear function. Unfortunately, they are usually **not transparent** or easily interpretable.



Efficacy vs Privacy

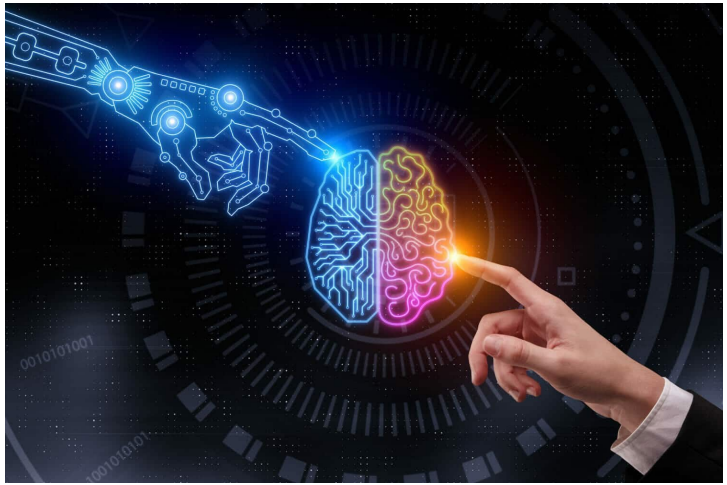
- Government regulatory frameworks such as the **GDPR** are enforcing data **privacy** as an inherent consideration in real-world systems.

- This requirement for **privacy** can adversely **limit explainability** in these systems.

Human Explanations vs Accuracy

- Even in the case where a model exhibits perfect transparency and we can easily observe the features that influence its decision-making ability, **it does not mean that the model is easily understandable to humans.**

- A trade-off exists between the **ability of a model to provide comprehensible explanations** and the **accuracy** of the model.



A brief history

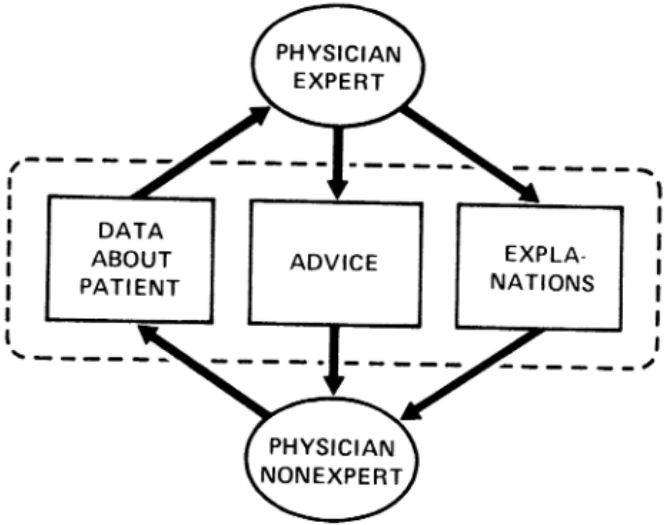
Artificial Intelligence

Explainable Artificial Intelligence

A brief history – Expert systems

- Beginning in the 1970s, computer scientists sought to develop models that could **emulate the decision-making of human experts** in a variety of fields.

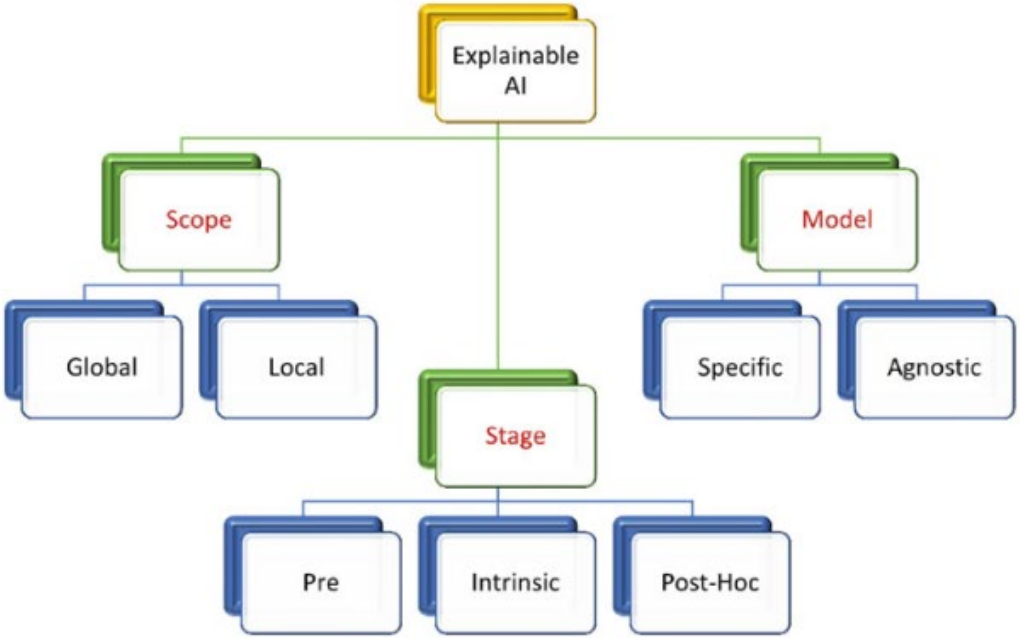
- **MYCIN** is a rule-based expert system and was developed to assist doctors in selecting appropriate antimicrobial therapies. This system provided both **medical advice** but also **explanations**.



Explanations in machine learning

- Most machine learning models work as **black boxes**.
- For this reason, explainability in **machine learning** is formulated as the problem **of finding an interpretable model** that **approximates** the black-box model.

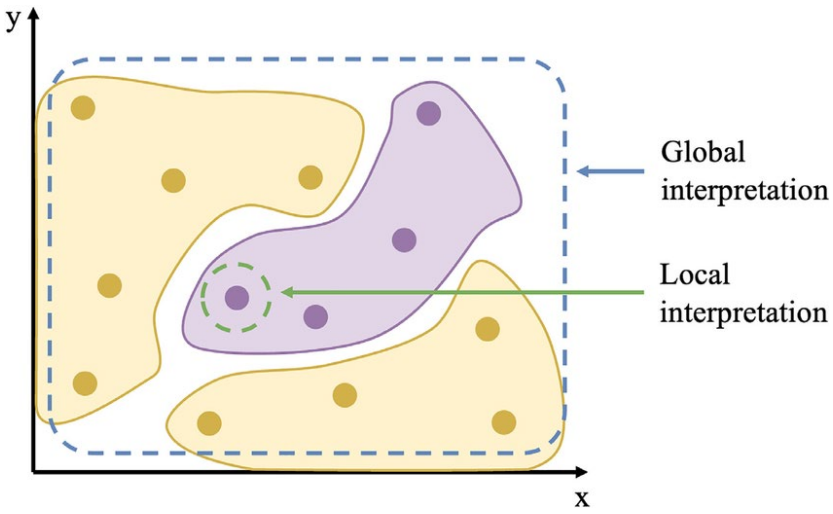
Explanation taxonomy



Scope of explanation

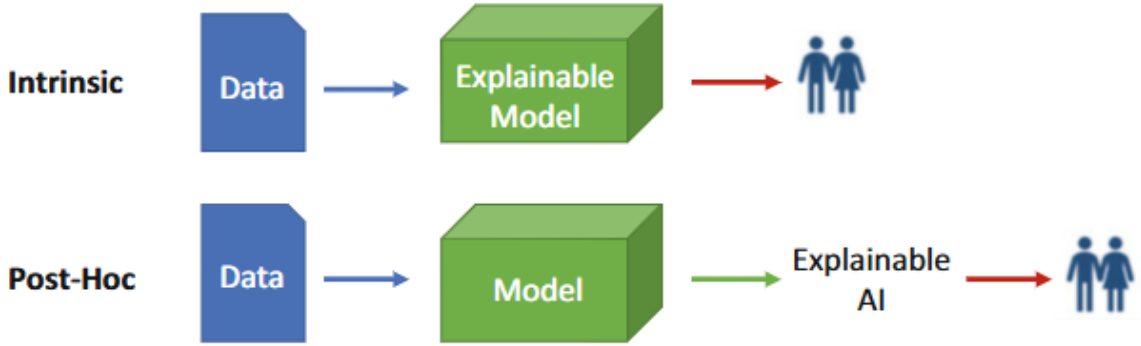
- **Global methods:** seek to explain the predictions of the overall model from a comprehensive, top-down approach.

- **Local methods** seek to explain how a specific sample is mapped to its output by providing us an understanding of how the model arrived at its prediction.



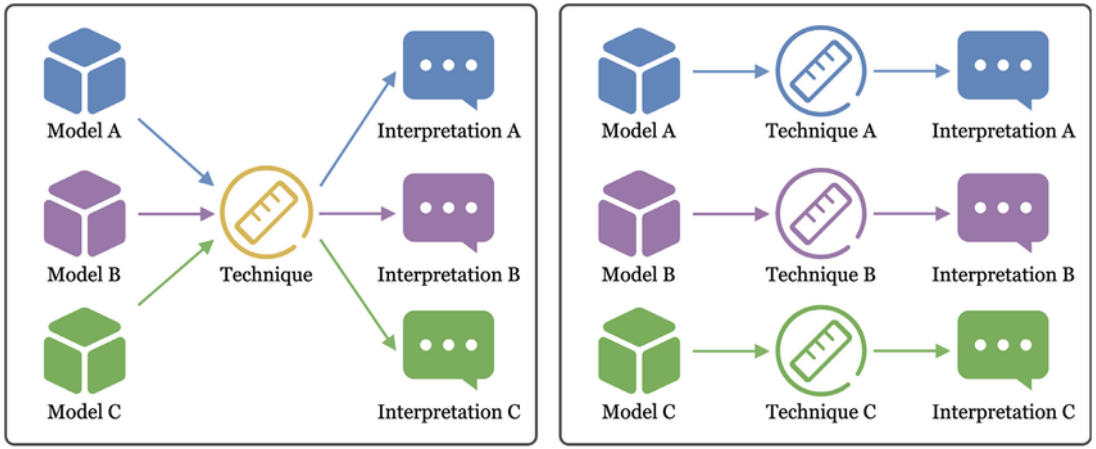
Intrinsic or post hoc?

- **Intrinsic interpretability** refers to machine learning models that are considered interpretable due to their simple structure, such as short decision trees or sparse linear models.
- **Post hoc** interpretability refers to the application of interpretation methods after model training.



Model-specific or model-agnostic?

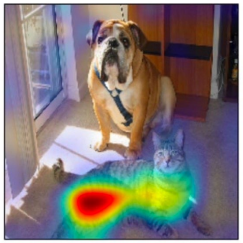
- **Model-specific** interpretation tools are limited to specific model classes.
- **Model-agnostic** tools can be used on any machine learning model and are applied after the model has been trained (post hoc).



(a) Model-agnostic (b) Model-specific

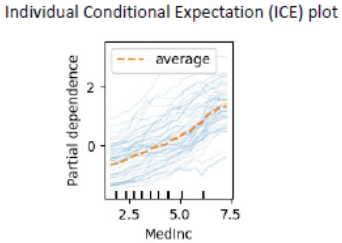
Explanation representation

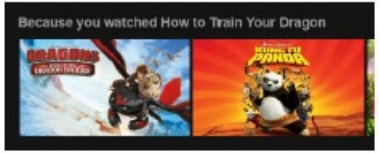
- **Feature importance**: indicate how much each feature contributed to the prediction for a given instance.
- **Local rules** provide insights into how a model behaves for a specific instance.
- **Visualizations:** use of visual representations to explain the behavior.
- **Explanations by example:** select or generate instances to explain.



(c) Grad-CAM 'Cat'

If Country is US, married, work hours > 45
→ Income>50K

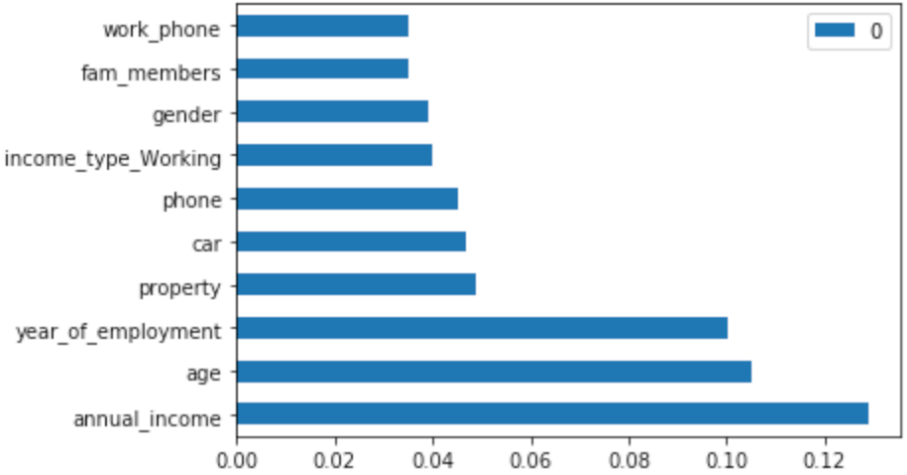




Feature importance

- Feature importance techniques aim to rank features based on how much they contribute to the prediction of a model
- These attributions can be presented **numerically, graphically, or in a tabular format** to show **which features are most influential** in driving the model's predictions.
- SHAP, Grad-CAM…

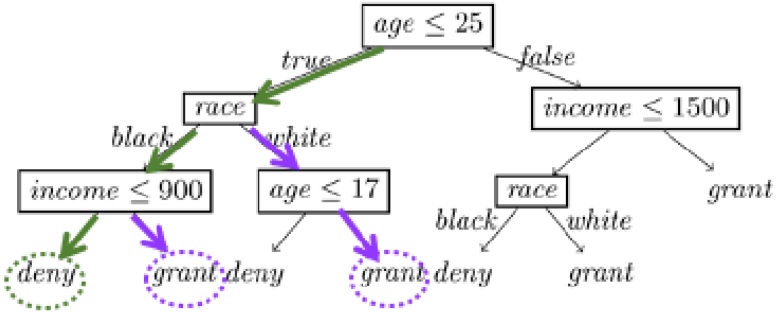
Example: In a credit risk model, feature importance might show that "credit score" and "income" are the most critical features for predicting whether a loan applicant will default.



Local rules

- **Local Rule-based Explanation** aims to extract a rule that provides a local explanation for a certain instance of the model. Often extracted from **local surrogate models**.
- They can be presented as logical statements, decision paths, or **decision rules** that describe the model's behavior for a particular input.
- LORE, ANCHOR…

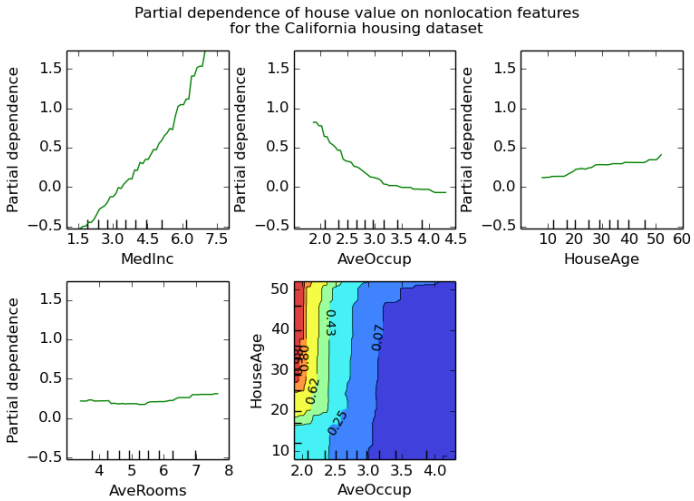
Example: the explanation of LORE for the denied request of a loan from a customer with "**age=22, race=black, and income=900**" to a bank that uses an AI could be the rule **if age ≤ 25 and race = black and income ≤ 900 then deny.**



Visualization-based explanations

- They use **visual representations** to explain the behavior
- These visual tools help explain model behavior, **highlight patterns**, and **show relationships** between inputs and outputs in an intuitive way.
- PDPs, ICE…

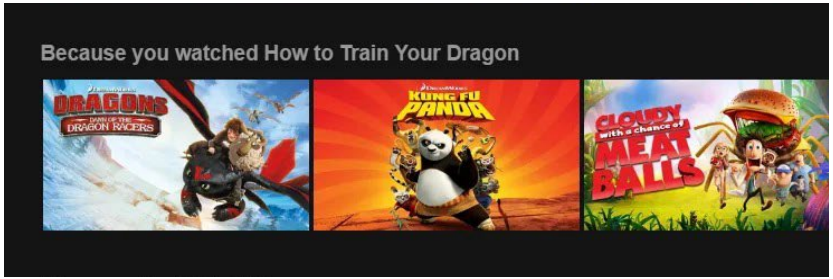
Example: For a house price prediction model, a partial dependence plot could show how the price varies with the size of the house, revealing the non-linear relationship between these variables.



Explanations by Example

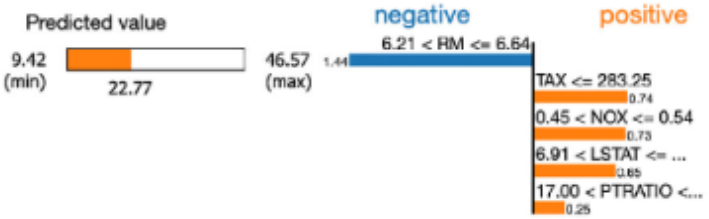
- Explanations by example involve **selecting or generating examples to help explain** a model's decision.
- Prototypes, Counterfactual Explanations…

Example: in recommendation systems, the model might present examples of similar content that the user watched, providing a comparison of what he would enjoy.



LIME (Local Interpretable Model-agnostic Explanations)

- Local, model agnostic explanation.
- **LIME** generates explanations by **creating surrogate models that are interpretable** and have a low number of features to keep the complexity of the interpretable model low.

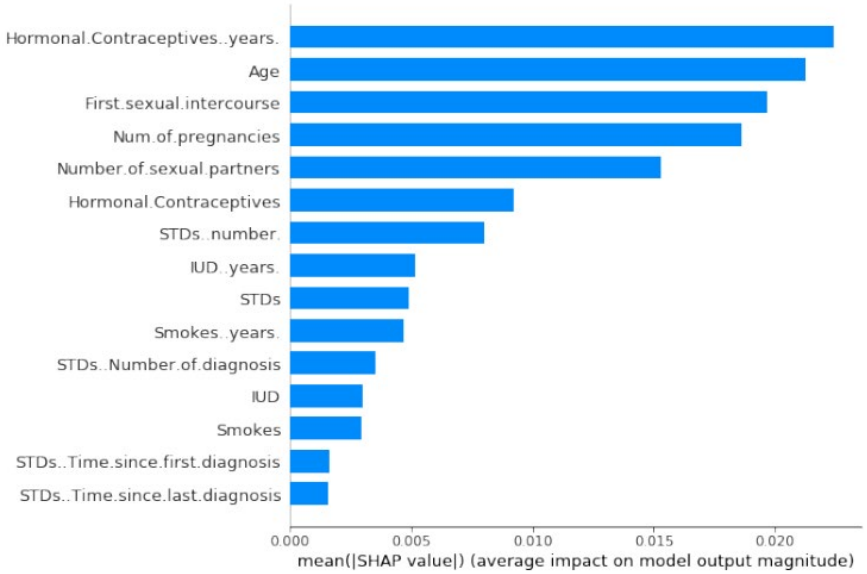


Feature	Value
RM	6.44
TAX	270.00
NOX	0.46
LSTAT	8.16
PTRATIO	18.20

```
[('6.21 < RM <= 6.64', -1.4447236495087157),
 ('TAX <= 283.25', 0.7417794586320381),
 ('0.45 < NOX <= 0.54', 0.7290587475071078),
 ('6.91 < LSTAT <= 11.38', 0.647686973013182),
 ('17.00 < PTRATIO <= 19.10', 0.2490023133440792)]
```

SHAP (Shapley Additive Explanations)

- Local, model agnostic explanation.
- **SHAP (Shapley Additive Explanations)** is a method based on **game theory** that explains the contribution of each feature to a model's prediction.



Online resources

There are excellent **online resources** with a collection of articles, books, tools, datasets, etc., all assembled in one place. Some of the links are:

1. https://github.com/jphall663/awesome-machine-learning-interpretability
2. https://github.com/lopusz/awesome-interpretable-machine-learning
3. https://github.com/pbiecek/xai_resources
4. https://github.com/h2oai/mli-resources
5. https://github.com/andreysharapov/xaience

References

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Thank you for your attention!

Any questions?

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